Monogenic Riesz Wavelet Representation for Micro-expression Recognition

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***Abstract*—A monogenic signal is a two-dimensional analytical signal that provides the local information of magnitude, phase, and orientation. While it has been applied on the field of face and expression recognition [1], [2], [3], there are no known usages for subtle facial micro-expressions. In this paper, we propose a feature representation method which succinctly captures these three low-level components at multiple scales. Riesz wavelet transform is employed to obtain multi-scale monogenic wavelets, which are formulated by quaternion representation. Instead of summing up the multi-scale monogenic representations, we consider all monogenic representations across multiple scales as individual features. For classification, two schemes were applied to integrate these multiple feature representations: a fusion-based method which combines the features efficiently and discriminately using the ultra-fast, optimized Multiple Kernel Learning (UFO-MKL) algorithm; and concatenation-based method where the features are combined into a single feature vector and classified by a linear SVM. Experiments carried out on a recent spontaneous micro-expression database demonstrated the capability of the proposed method in outperforming the state-of-the-art monogenic signal approach to solving the micro-expression recognition problem.**

***Keywords*—Monogenic signal; Riesz wavelet transform; facial micro-expressions; quaternion representation; UFO-MKL.**

I. Introduction

A micro-expression is actually a normal facial expression that occurs very rapidly (typically from 1/25 to 1/5 of a second) [4]. It expresses one of the six universal expressions, such as happy, sad, fear, surprise, anger, and disgust. Micro-expressions are brief and involuntary, the latter usually reveals the genuine emotion state of a person who is trying to conceal his/her emotions [5] and thus it is hard for people to identify and recognize the micro-expression with the naked eye [6]. The recognition difficulty was quantified in experiments conducted by Frank et al. [7]. The results revealed that the accuracy rate of recognizing five subtle expressions were 32% and 47% for untrained and trained human experts respectively.

In recent years, micro-expression recognition and its analysis has raised much interest in psychological research. But in image and signal processings, little attention was paid to micro-expression recognition. This might due to the lack of well-established databases and the nature of micro-expressions which have short duration and low intensity. To date, there are two publicly available spontaneous micro-expression databases: CASMEII [8] and SMIC [9]. An evaluation protocol for micro-expression recognition was proposed in [8] whereby the publicly available CASME II spontaneous facial micro-expression database was introduced. In their paper, the baseline methods used are a block-based Local Binary Pattern-Three Orthogonal Planes (LBP-TOP) [10] with 5×5 partitions for feature extraction and Support Vector Machine (SVM) for classification based on leave-one-video-out cross-validation. In SMIC database, an evaluation protocol with LBP-TOP (as feature extractor) and SVM (as classifier) based on leave-one-subject-out cross-validation (LOSO) was introduced. In their work, temporal interpolation model (TIM) was applied to vary (up-sampling and down-sampling) the frame numbers to promote the feature extraction process. More recently, some works [11], [12] applied the optical strain technique to extract the relatively small amount of deformation of a face in a video sequence.

In this paper, we present a new feature representation method which is capable of capturing low-level monogenic components from multiple scales. To extract features, a two-layer architecture as proposed in the Improved Spatiotemporal Local Monogenic Binary Pattern (ISTLMBP) method [13] for emotion recognition "in-the-wild" is adopted to describe the video sequences. In the first layer, Riesz wavelet transform is applied on input images to extract the multi-scale monogenic wavelets for capturing the low-level features, i.e., magnitude, phase and orientation. The local phase at a point can be positive or negative depending on the orientation that is expressed by an angle δ or its antipode $\delta + \pi$. To solve the phase ambiguity problem, multi-scale monogenic wavelets are formulated by quaternion representation. The scalar values of the magnitude, phase and orientation for each scale are then computed. In the second layer, these low-level features are encoded and histograms are constructed. The UFO-MKL method [14] and the concatenating method are used to fuse multiple feature vectors and fed into classifier.

There are four main differences of this work compared to that of ISTLMBP [13]: (1) Wavelet transform is employed to form multi-scale signal, vs log-gabor filters are used in ISTLMBP; (2) Quaternion phase representation is used to avoid the ambiguous sign of phase components, vs scalar phase is used in ISTMBP; (3) Low-level feature components (i.e. magnitude, phase and orientation) at multiple scales are utilized, vs those low-level feature components from sum of multi-scale representation are used in ISTLMBP; and (4) both SVM and MKL are adopted as classifiers, vs solely MKL classifier is used in ISTLMBP.

This paper is organized as follows: Section 2 gives a brief review of related literature, followed by Section 3 that describes the proposed method in detail. Next, Section 4

summarizes the experimental results while conclusions are presented in Section 5.

II. Related work

An analytical signal [15] is a complex-valued continuous function which covers the whole spectrum of the signal without using negative frequencies. A monogenic signal [16] is an isotropic 2D extension of an analytical signal, in which the local rotational-invariant properties of the signals i.e. magnitude, phase and orientation that correspond to energetic, structural and geometric information respectively, can be extracted. Monogenic representation has been recently demonstrated in several applications such as analysis of medical data [17], face recognition [2], facial expression recognition [3] etc.

Local Binary Pattern (LBP) [18] is a feature extractor widely used in computer vision as it is a simple yet powerful texture descriptor. It encodes relative comparison between intensity of a central pixel and its neighborhood into a binary sequence and statistically summarizes these sequences through their histogram. Block-based LBP-TOP [10] is the spatio-temporal extension of LBP as it considers the texture from three orthogonal planes XY, XT and YT where X, Y, T are horizontal, vertical and temporal axes respectively. Moreover, the block-based approach divides input images into several blocks before LBP-TOP histograms are extracted from block volumes and concatenated into a single column vector. Yang et al. [2] extended it to Monogenic Binary Pattern (MBP) by encoding the magnitude and orientation signals separately which are later also concatenated into a single feature vector.

Given that information is extracted from different monogenic components, it is important to select the significant information by learning model parameters based on an optimal kernel combination. Multiple Kernel Learning (MKL) [19] is used when there are heterogeneous representations of data for the classification task. It is able to construct a kernel model where the kernel is a linear combination of fixed base kernels and the data is weighted using different weighting coefficients. Recently, UFO-MKL [14] (Ultra Fast Optimization-MKL) is proposed, with a much shorter computation time and the level of sparsity is adjustable depending on the characteristics of the input data.

There are many facial expression databases that are well established. On the contrary, only limited micro-expression databases are available due to difficulties in capturing the subtle expression changes and labeling of data. To date, all the existing micro-expressions databases are recorded under controlled lab conditions. However, there are known drawbacks in the existing micro-expression databases: micro-expressions being largely posed instead of spontaneous [20], [21], and the insufficiency of sample videos for certain subjects [9].

III. Proposed method

The flowchart of the proposed approach is illustrated in Fig. 1. Concisely, we propose the following steps to recognize micro-expressions automatically: (1) Transform input images into multi-scale monogenic wavelets by Riesz wavelet transform; (2) Turn the multi-scale monogenic wavelets into quaternion representation; (3) Extract the scalar value of magnitude, phase and orientation (both real and imaginary) from each scale from the quaternionic representations; (4) Apply feature extraction techniques onto the low-level features (magnitude, phase and orientation) at each scale and encode them into histograms. The respective feature extraction methods and encoding schemes are described in Section III-B; (5) Finally, evaluate the proposed method using UFO-MKL [14] and linear SVM for decision-level features fusion approach and feature-level features fusion approach respectively.

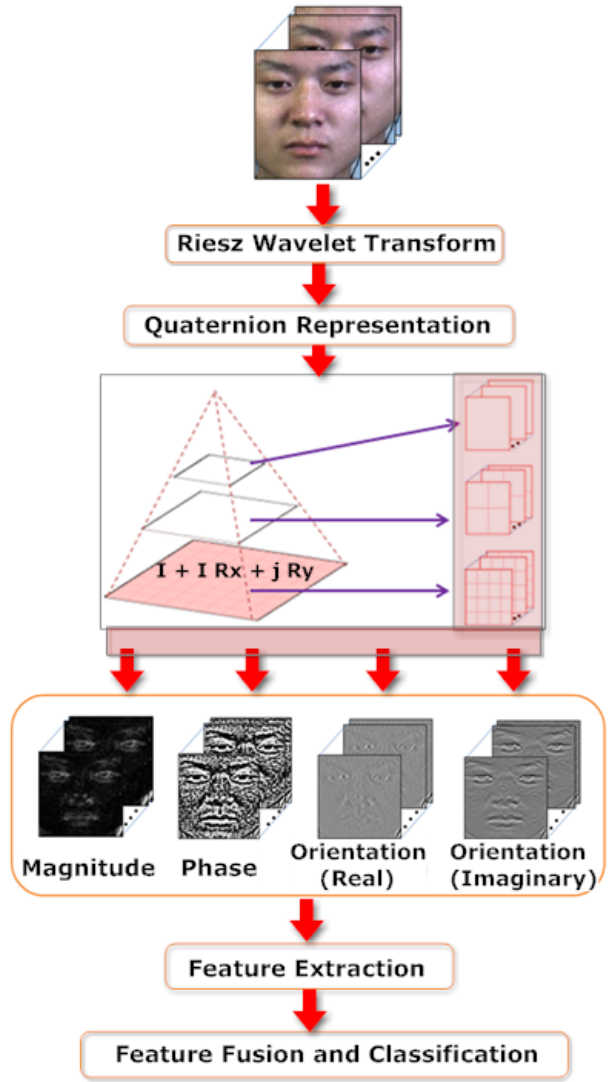

Fig. 1: Flowchart of the proposed algorithm

A. 2D Riesz Wavelet Pyramid

For an image $I(x, y)$, the multi-scale monogenic signal as shown in Eq. 1 can be obtained by applying 2D monogenic analysis to each scale of the 2D isotropic wavelet pyramid. In short, the Riesz transform is applied onto the wavelet pyramid to form the multi-scale monogenic wavelets:

$$f_{mono,k}(x) = (\psi_k * f(x), R_1\{\psi_k * f\}(x), R_2\{\psi_k * f\}(x)) \quad (1)$$

where $f(x)$ is the input signal,
$x = (x_1, x_2) \in \mathbb{R}^2$,

R_i is the Riesz operator,
ψ_k is the wavelet function and
$k = 1...K$, K is the number of wavelet scale.

To solve the ambiguous sign of the local phase component as the local phase at a particular point can be either positive or negative depending on whether the orientation is expressed by an angle δ or its antipode $\delta + \pi$, the Riesz wavelet triples are converted to the quaternionic formulation [22]:

$$\mathbf{r} = I + iR_x + jR_y \tag{2}$$

where $I = \psi_k * f(x)$,
$R_x = R_1\{\psi_k * f\}(x)$,
$R_y = R_2\{\psi_k * f\}(x)$.
Then, the quaternion amplitude and quaternion phase can be expressed as:

$$\begin{aligned} A_{quaternion} &= \|\mathbf{r}\| \\ \phi_{quaternion} &= \log(\mathbf{r}/\|\mathbf{r}\|) \end{aligned} \tag{3}$$

By quaternionic formulation, the three important components of monogenic signal— magnitude A, filtered phase ϕ and orientation θ can be calculated in scalar value as follows:

$$A_{\mathbf{r}} = \sqrt{I^2 + R_x^2 + R_y^2} \tag{4}$$

$$\phi_{\mathbf{r}} = arctan(\sqrt{R_x^2 + R_y^2}/I) \tag{5}$$

$$\theta_{\mathbf{r}} = arctan(R_y/R_x) \tag{6}$$

B. Feature extraction

We employ the feature encoding techniques in [13] to encode the monogenic components. In this step, the magnitude, phase, and the real and imaginary orientations of face images are collected sequentially to form 3-D matrices for each scale. As such, the appearance, horizontal and vertical motion (akin to information from the three orthogonal planes XY, XT and YT) in each image for each scale can be fully utilized to enrich the features. To extract the local texture pattern, block-based method is implemented, where the image is first partitioned into $N \times N$ non-overlapping blocks, then the histogram is obtained from each block volume through the encoding schemes described hereafter.

1) Magnitude: The uniform LBP operator encodes the magnitude details of the local variation energy from three orthogonal planes, using a thresholding technique by comparing the center pixel to the neighboring pixels:

$$\begin{aligned} H_A(x_c, y_c) &= \sum_{p=0}^{P-1} s(A_{x_p,y_p} - A_{x_c,y_c})2^P, \\ s(A_{x_p,y_p} - A_{x_c,y_c}) &= \begin{cases} 1, & A_{x_p,y_p} - A_{x_c,y_c} \geq 0 \\ 0, & A_{x_p,y_p} - A_{x_c,y_c} < 0 \end{cases} \end{aligned} \tag{7}$$

where P is the number of neighboring points around the center pixel (x_c, y_c), A_{x_c,y_c} and A_{x_p,y_p} are the monogenic magnitudes of the center pixel and neighboring points (equally spaced on a circle of fixed radius) respectively.

2) Orientation: The monogenic orientation contains real and imaginary elements and they can be encoded as follows:

$$B(R_i) = \begin{cases} 1, & R_i > 0; \\ 0, & R_i \leq 0. \end{cases} \tag{8}$$

where $i = x$ and $i = y$ for the real and imaginary pictures respectively. Local bit exclusive-OR operator (LXP) is used to compute the correlation between the center pixel and the neighboring pixels from three orthogonal planes. The histogram of real or imaginary pictures can be denoted as:

$$H_\theta = \sum_{p=0}^{P-1} (B(f_{x_p,y_p}) \oplus B(f_{x_c,y_c}))2^p \tag{9}$$

where $B(f_{x_p,y_p})$ is the Phase-Quadrant Demodulation Coding (PQDC) value of P neighboring points around center pixel (x_c, y_c). $\oplus$ indicates the bit exclusive-OR operator.

3) Phase: The monogenic phase information is encoded by quantifying the phase angle with the function:

$$q = mod([\frac{\phi(x,y)}{\frac{2\pi}{\varphi}} + 0.5], \varphi) \tag{10}$$

where $\phi(x, y)$ denotes the monogenic phase angle, and φ is the number of dominant orientations. The dominant phase bins, which are the neighboring angles $\phi(x, y)$ and $q_(x, y)$ from three orthogonal planes can easily obtained by:

$$Q_p = \begin{cases} 0, & q_{x,y} = q_{x_p,y_p} \\ 1, & q_{x,y} \neq q_{x_p,y_p} \end{cases} \tag{11}$$

Finally, the phase histogram is computed as follows:

$$H_\phi = \sum_{p=0}^{P-1} Q_p 2^p \tag{12}$$

C. Feature fusion and classification

To fuse the multiple feature representations across all scales, two approaches were employed – MKL (used in the original paper [13] based on UFO-MKL implementation [14]) and normal concatenation. The MKL approach fuses all three feature components across K scales. For each k-th scale, its corresponding features $H_{k,c}$, where $c = \{A, \theta, \phi\}$, $k \in [1, K]$ are then kernelized into orthogonal matrices of size $D_s \times D_s$.

The monogenic Riesz wavelet pyramid (based on a dyadic scale space) results in different feature dimensions D_s for each scale s, i.e. the dimension sizes are halved with the increase in scale. Hence, we opt to concatenate them on each scale instead of summing the scalar values across all scales in [13]. We argue that the latter may potentially result in redundancy of non-discriminating signal information. Meanwhile, the SVM classifier straightforwardly takes in a single feature vector by concatenating all features $H_{k,c}$ directly. Linear kernels are used instead of Gaussian or higher-order kernels to avoid over-fitting our data, which is much smaller in sample size.

IV. Experiment

The proposed algorithm was evaluated on the most recent spontaneous micro-expression database, CASME II [8]. In our experiment, 247 video sequences elicited from 26 subjects, consists of 5 types of micro-expressions: happiness, disgust,

surprise, repression and tense. Videos were collected using Point Grey GRAS-03K2C camera at a resolution of 280×340 and a high-speed frame rate of 200 *fps*. The average length of each video is approximately 0.34 seconds, while the longest and shortest sequence are 146 and 24 frames respectively.

A. Experiment settings

We compared our proposed method against two state-of-the-art approaches. First, we evaluated the baseline method reported in the original database paper [8]. Block-based LBP-TOP (uniform patterns) was used for feature extraction. We set the radii of the (XY, XT, YT) planes to (1,1,4) while the number of neighbor points, $P = 4$. In the classification stage, SVM with linear kernel was employed under leave-one-subject-out cross-validation (LOSOCV) setting. For best results, we used $c = 10000$.

We also benchmark our method against the recently proposed Improved STLMBP method [13] which is formulated based on monogenic signals. A Poisson-scale filter with scale number of 3 (equivalent to $K = 3$ for our case) was applied to input frames to obtain multi-scale representation. All the component scalars across all scales are summing up, as done in the original paper. For the proposed method, Riesz wavelet transform is applied to produce multi-scale monogenic wavelets. We fix the wavelet pyramid scale to $K = 3$, resulting in three different dimensions of monogenic wavelets (i.e., $120 \times 120, 60 \times 60$ and 30×30 in this case).

For all low-level features (magnitude, orientation and phase), the radii of the (XY, XT, YT) planes used were set to $(1, 1, 4)$, and the number of neighbor points in each plane is set to 4. Uniform patterns are used in the construction of the histogram. In order to have a consistent feature length, a fixed blocking size must be applied to the component images of each scale.

In all our experiments, input image frames are resized to 120×120 pixels. We also used two block partitioning sizes, $N = 4$ and $N = 5$ during feature extraction, as these are empirically found to perform the best for CASME II sample resolution. For the monogenic-based methods, the number of dominant phase orientations is fixed to $\phi = 4$. Also, MKL and SVM are applied for classification, with linear kernels ($c = 10000$). Due to iterative convergent nature of MKL, we repeat their experiments for 100 times and take the averaged results. As the database has multiple subjects, LOSOCV setting is applied in our evaluation, whereby for each fold, the image sequences of one subject are hold-out as testing data and the remaining images are used for training the classifier. This process is repeated for J times, for J subjects.

B. Results and discussions

For clarity, the abbreviations MBP (for Improved STLMBP) and RW (for our proposed Monogenic Riesz Wavelet) are used in both Fig. 2 and Table I. The recognition accuracy for all evaluated methods (Improved STLMBP [13] and our proposed method on both MKL and SVM, and baseline LBP-TOP from[8]) on CASME II database are shown in Fig. 2.

As the CASME II database is highly imbalanced (varying number of video samples for each class), with about 40% of samples belonging to the "tense" class, while the rest are disproportionally distributed as well. Thus, we also compute the F1-measure, recall and precision metrics, which are suited for imbalanced datasets [23]. Briefly, recall is the ratio of relevant data that are retrieved from the database, precision is the ratio of retrieved data that are relevant in the database, while F1-measure is the harmonic mean of precision and recall. In Table I, the F1-measure of the proposed method remains competitive, if not surpassing a majority of the evaluated methods. In particularly, the RW+SVM combination is thoroughly superior in 5×5 blocks. In 4×4 blocks setting, the RW methods trail the LBP-TOP baseline slightly in terms of the precision-recall balance, although it seemed to be more precise in retrieval.

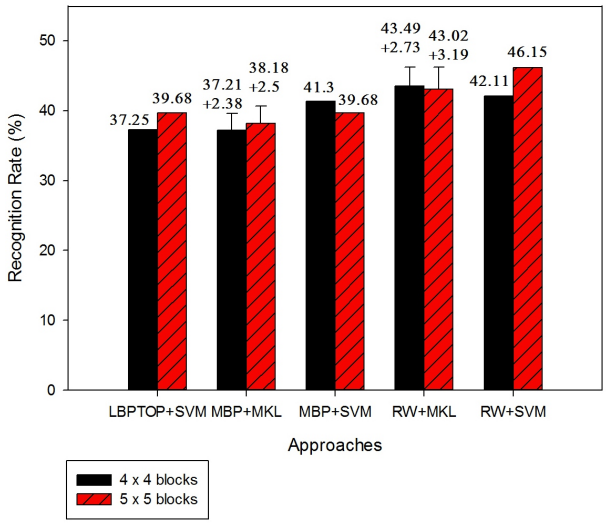

Fig. 2: Subtle emotion recognition accuracy rate w.r.t different approaches. The ticks above the MKL methods indicate the upper bound level (plus standard deviation).

TABLE I: Comparison of F1-measure, recall and precision of different approaches

Blocks	Approaches	F1	Recall	Precision
4x4	LBPTOP+SVM	0.3431	0.3372	0.3493
	MBP+MKL	0.2995	0.2993	0.2998
	MBP+SVM	0.3255	0.3210	0.3301
	RW+MKL	0.3224	0.3033	0.3462
	RW+SVM	0.3252	0.2997	0.3554
5x5	LBPTOP+SVM	0.3715	0.3558	0.3887
	MBP+MKL	0.2946	0.2831	0.3075
	MBP+SVM	0.3136	0.2989	0.3299
	RW+MKL	0.3216	0.2851	0.3781
	RW+SVM	0.4307	0.3540	0.5497

V. Conclusion

In this paper, we propose a spatio-temporal feature representation approach for facial micro-expression recognition in video based on monogenic signals. For each video sample, the three low-level monogenic features, namely magnitude, phase and orientation information are captured from multiple scales using Riesz wavelet transform. To provide invariance

towards the ambiguity of phase sign, quaternionic phase is formed by converting the monogenic wavelets to quaternion representation. In the classification step, the scalar values of the three monogenic components from each scale are computed and treated as individual features instead of summing them up to avoid loss of information. Experiments conducted on the recent CASME II micro-expression database showed that the proposed Monogenic Riesz Wavelet method is able to achieve significant improvement over the baseline LBP-TOP and state-of-the-art Improved STLMBP methods.

In future, an effective way of dealing with noise from the phase and orientation feature components can be further explored for added improvement. Also, we intend to discover the optimum number of dominant orientations for micro-expressions in contrast to that of normal facial expressions.

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